EE1-10B MATHEMATICS II

1. a) Given the equations of three planes

$$\underline{\mathbf{r}}\cdot(1,-1,2)=2\,,\qquad \underline{\mathbf{r}}\cdot(0,1,-3)=\alpha\,,\qquad \underline{\mathbf{r}}\cdot(2,1,-5)=1$$

show that when $\alpha=1$ the three planes do not intersect, but form the sides of a prism. Find the value of α so that the three planes intersect, and obtain the intersection.

Solution:

Can eliminate variables from equations, but more elegant to use Gaussian elimination. Each plane equation gives one row of the augmented coefficient matrix:

$$(A:\underline{b}) = \begin{pmatrix} 1 & -1 & 2 & 2 \\ 0 & 1 & -3 & \alpha \\ 2 & 1 & -5 & 1 \end{pmatrix} \overset{R_3-2R_1}{\sim} \begin{pmatrix} 1 & -1 & 2 & 2 \\ 0 & 1 & -3 & \alpha \\ 0 & 3 & -9 & -3 \end{pmatrix}$$
$$\overset{R_3-3R_2}{\sim} \begin{pmatrix} 1 & -1 & 2 & 2 \\ 0 & 1 & -3 & \alpha \\ 0 & 0 & 0 & -3-3\alpha \end{pmatrix}$$

Hence if $\alpha=1$, row 3 represents $0x+0y+0z=-6$, and the equations are inconsistent: no intersection exists. None of the planes are parallel, as none of the normal vectors are multiples of each other: the only possibility left is that the planes form the sides of a prism.

Many people forgot to mention this: it's a prism only if none of the planes are parallel to each other. Most people took the approach of making $\alpha=1$, then solving the system to show inconstistency, without getting information on the required value of α for consistency. Of these, many had trouble manipulating the matrix with an unknown, resulting in arithmetic errors.

For an intersection to exist, require consistent equations: $\alpha=-1$. Row 2 then gives $y-3z=-1$.

Choose the free variable $z=\lambda\in\mathbb{R}\Rightarrow y=-1+3\lambda$ and then Row 1 gives $x=2+y-2z=2+(-1+3\lambda)-2\lambda=1-\lambda$. In vector form:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix},$$

the equation of a line in $\mathbb{R}^3$¿

[6]

A considerable minority gave the solution as a single point.

b) Show that for any three vectors $\underline{\mathbf{u}}, \underline{\mathbf{v}}, \underline{\mathbf{w}}$: $[(\underline{\mathbf{u}}+\underline{\mathbf{v}})\times(\underline{\mathbf{v}}-\underline{\mathbf{w}})]\cdot(\underline{\mathbf{u}}+\underline{\mathbf{w}}) = 0$.

Solution:

Begin with $(\underline{\mathbf{u}}+\underline{\mathbf{v}})\times(\underline{\mathbf{v}}-\underline{\mathbf{w}}) = \underline{\mathbf{u}}\times\underline{\mathbf{v}} - \underline{\mathbf{u}}\times\underline{\mathbf{w}} - \underline{\mathbf{v}}\times\underline{\mathbf{w}}$, as $\underline{\mathbf{v}}\times\underline{\mathbf{v}} = \underline{0}$ Hence we have

$$(\underline{\mathbf{u}}\times\underline{\mathbf{v}} - \underline{\mathbf{u}}\times\underline{\mathbf{w}} - \underline{\mathbf{v}}\times\underline{\mathbf{w}})\cdot\underline{\mathbf{u}} + (\underline{\mathbf{u}}\times\underline{\mathbf{v}} - \underline{\mathbf{u}}\times\underline{\mathbf{w}} - \underline{\mathbf{v}}\times\underline{\mathbf{w}})\cdot\underline{\mathbf{w}}$$
$$= -(\underline{\mathbf{v}}\times\underline{\mathbf{w}})\cdot\underline{\mathbf{u}} + (\underline{\mathbf{u}}\times\underline{\mathbf{v}})\cdot\underline{\mathbf{w}},$$

as all the other terms are zero, e.g. $(\underline{\mathbf{v}}\times\underline{\mathbf{w}})\cdot\underline{\mathbf{v}} = 0$, etc. Rearrange:

$$= -\underline{\mathbf{u}}\cdot(\underline{\mathbf{v}}\times\underline{\mathbf{w}}) + \underline{\mathbf{w}}\cdot(\underline{\mathbf{u}}\times\underline{\mathbf{v}})$$

Finally, the triple scalar product gives that $\underline{\mathbf{u}}\cdot(\underline{\mathbf{v}}\times\underline{\mathbf{w}}) = \underline{\mathbf{w}}\cdot(\underline{\mathbf{u}}\times\underline{\mathbf{v}})$ and the result follows.

[5]

Few students took the easy approach outlined above. Most wrote everything out in terms of components $u_1, u_2, u_3, v_1, v_2, \ldots$ which led to an intractable mess of terms $u_i v_j w_k$, often with errors.

c) Consider the matrix

$$A = \begin{pmatrix} 1 & -2 & 2 \\ 0 & -1 & 2 \\ -6 & 5 & -4 \end{pmatrix}.$$

i) Show that $\lambda = 1$ is an eigenvalue of A, and find the other eigenvalues.

Solution:

Set up

$$\det(A-\lambda I) = \begin{vmatrix} 1-\lambda & -2 & 2 \\ 0 & -1-\lambda & 2 \\ -6 & 5 & -4-\lambda \end{vmatrix}$$

and expand by first column:

$$= (1-\lambda)[(-1-\lambda)(-4-\lambda) - 10] - 6[-4-2(-1-\lambda)]$$

Note $(1-\lambda)$ in first term, and don't multiply out:

$$= (1-\lambda)(\lambda^2+5\lambda-6) + 12(1-\lambda) = (1-\lambda)(\lambda^2+5\lambda+6) = 0$$

Hence $\lambda = 1$ is an eigenvalue and the quadratic term factorizes to give $\lambda = -2, -3$ as the other eigenvalues.

[4]

Most people got this right, well done. Quite a sizable minority had problems factorizing. Of those, a few simply wrote down $det(A - 1\cdot I)$ which is zero, thereby showing that $\lambda = 1$ is indeed an eigenvalue, which gave at least a partial solution.

ii) Find an eigenvector of A, corresponding to $\lambda = 1$.

Find $\underline{\mathbf{x}} = (x, y, z)^T$ such that $(A - 1I)\underline{\mathbf{x}} = \underline{0}$:

$\begin{pmatrix} 0 & -2 & 2 \\ 0 & -2 & 2 \\ -6 & 5 & -5 \end{pmatrix}\underline{\mathbf{x}} = \underline{0}$ and no further operations are required. Row 1 and 2 both give $y = z$. Row 3 gives $-6x + 5y - 5z = 0 \Rightarrow x = 0$ and

an eigenvector for $\lambda = 1$ is $(0,1,1)^T$.

[3]

Again, many good answers, but a sizable minority got as far as the matrix $A - I$ and didn't know where to go from here. Many others intrepreted the equation in rows 1 and 2 as $x = y$ or $y = -z$.

d) Let A be an invertible matrix, and λ an eigenvalue of A.

i) Show that $\lambda \neq 0$;

If $\lambda = 0$ were an eigenvalue of A, then $A\underline{\mathbf{x}} = 0\underline{\mathbf{x}}$ for a non-zero eigenvector $\underline{\mathbf{x}}$. Hence $A\underline{\mathbf{x}} = \underline{0}$ would have a nonzero solution. But if A is invertible, the only solution for $A\underline{\mathbf{x}} = \underline{0}$ is $\underline{\mathbf{x}} = \underline{0}$. Hence zero cannot be an eigenvalue, and $\lambda \neq 0$

[3]

Lots of arguments, including diffrerent valid approaches: well done! Many attempted to show things about $det(A - \lambda I)$, claiming this is equal to $detA - \lambda det(I)$, which it is not.

ii) Hence, or otherwise, show that $1/\lambda$ is an eigenvalue of A^{-1}.

Given that $\lambda \neq 0$ is an eigenvalue, we have non-zero $\underline{\mathbf{x}}$ such that $A\underline{\mathbf{x}} = \lambda\underline{\mathbf{x}}$. As A^{-1} exists, we can multiply on left by A^{-1}:

$A^{-1}(A\underline{\mathbf{x}}) = A^{-1}(\lambda\underline{\mathbf{x}}) \Rightarrow (A^{-1}A)\underline{\mathbf{x}} = \lambda A^{-1}\underline{\mathbf{x}} \Rightarrow \underline{\mathbf{x}} = \lambda A^{-1}\underline{\mathbf{x}}$.

Now divide by $\lambda \neq 0$ to obtain $\frac{1}{\lambda}\underline{\mathbf{x}} = A^{-1}\underline{\mathbf{x}}$ and as $\underline{\mathbf{x}}$ is non-zero, we see that $1/\lambda$ is an eigenvalue for A^{-1} .

[4] *Most got this right, but many did not justify their deductions with enough detail. You need to clarify why something works. The above is enough detail.*

2. a) i) Evaluate the determinant of the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 2 \\ 2 & -1 & 1 \\ \alpha & 1 & -5 \end{pmatrix}$$

and state the value of α for which $\mathbf{A}$ is singular.

Solution:

Expand by first column:

$\det(A) = 1(5-1) - 2(5-2) + \alpha(-1+2) = -2 + \alpha$.

So the matrix is singular, $\det(A) = 0$ when $\alpha = 2$.

[3]

Most people got this right, a few arithmetic errors.

ii) Let $\alpha = 3$. Use Gauss-Jordan elimination (row operations) to find $\mathbf{A^{-1}}$.

Solution:

Set up the augmented matrix, and use row operations, one column at at time, left to right:

$$
\begin{aligned}
(A:I) \quad &= \quad \left(\begin{array}{ccc|ccc} 1 & -1 & 2 & 1 & 0 & 0 \\ 2 & -1 & 1 & 0 & 1 & 0 \\ 3 & 1 & -5 & 0 & 0 & 1 \end{array}\right) \\
&\overset{\substack{R_2-2R_1 \\ R_3-3R_1}}{\sim} \quad \left(\begin{array}{ccc|ccc} 1 & -1 & 2 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 4 & -11 & -3 & 0 & 1 \end{array}\right) \\
&\overset{\substack{R_1+R_2 \\ R_3-4R_2}}{\sim} \quad \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & -1 & 1 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -4 & 1 \end{array}\right) \\
&\overset{\substack{R_1+R_3 \\ R_2+3R_3}}{\sim} \quad \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 4 & -3 & 1 \\ 0 & 1 & 0 & 13 & -11 & 3 \\ 0 & 0 & 1 & 5 & -4 & 1 \end{array}\right)
\end{aligned}
$$

So that $A^{-1} = \begin{pmatrix} 4 & -3 & 1 \\ 13 & -11 & 3 \\ 5 & -4 & 1 \end{pmatrix}$.

[5]

Lots of good answers. Quite a lot of arithmetic errors. A few people solved it using cofactors and the adjoint matrix.

iii) Use the inverse found in (ii) to solve the set of linear equations

$$
\begin{array}{rcrcrcr} -x & + & 2y & + & z & = & 3 \\ -x & + & y & + & 2z & = & 0 \\ x & + & 3y & - & 5z & = & -8 \end{array}
$$

Solution:

Spot the rearrangement of the equations:

$$
\begin{array}{rcrcrcr} -x & + & y & + & 2z & = & 0 \\ -x & + & 2y & + & z & = & 3 \\ x & + & 3y & - & 5z & = & -8 \end{array}
$$

and then

$$
\begin{array}{rcrcrcr} y & - & x & + & 2z & = & 0 \\ 2y & - & x & + & z & = & 3 \\ 3y & + & x & - & 5z & = & -8 \end{array}
$$

(can be done in one step!)

so the system is equivalent to $A\underline{\mathbf{x}} = (0,3,-8)^T$ where $\underline{\mathbf{x}} = (y,x,z)^T$.

Now use the inverse, $\underline{\mathbf{x}} = A^{-1}(0,3,-8)^T$:

$$
\begin{pmatrix} y \\ x \\ z \end{pmatrix} = \begin{pmatrix} 4 & -3 & 1 \\ 13 & -11 & 3 \\ 5 & -4 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 3 \\ -8 \end{pmatrix} = \begin{pmatrix} -17 \\ -57 \\ -20 \end{pmatrix}
$$

so the solution is $x = -57$, $y = -17$ and $z = -20$.

[4]

Many, many people failed to see the necessary rearrangements, applying A^{-1} to the vector as given. Or saw one but not the other. Or saw that a rearrangement is necessary, but applied it to the inverse matrix instead, all of which led to errors. A number of people couldn't see how to apply the rearrangement, clearly gave up after a few attempts and then solved the equations directly, either by A-level elimination of variables, or by Gaussian elimination. Interestingly, a recurrent error, independent of any of the above was that most people, not just a few, but more than half of everyone wrote down the vector $(3,0,-8)$ at the beginning and later on switched the -8 to 8. Strikingly odd!

b) Given the function $f(x) = \dfrac{1}{\sqrt{1-x}}$,

i) Obtain the Maclaurin series for $f(x)$ up to the term in x^3 and state the remainder term

Solution:

$$\begin{aligned} f(x) &= (1-x)^{-1/2} &\Rightarrow\quad f(0) &= 1 \\ f'(x) &= -\frac{1}{2}(1-x)^{-3/2}(-1) &\Rightarrow\quad f'(0) &= \frac{1}{2} \\ f''(x) &= \frac{3}{4}(1-x)^{-5/2} &\Rightarrow\quad f''(0) &= \frac{3}{4} \\ f'''(x) &= \frac{15}{8}(1-x)^{-7/2} &\Rightarrow\quad f'''(0) &= \frac{15}{8} \\ f^{(4)}(x) &= \frac{105}{16}(1-x)^{-9/2} \end{aligned}$$

So the Maclaurin series is

$$\begin{aligned} (1-x)^{-1/2} &= 1+\frac{1}{2}x+\frac{3}{4\cdot 2!}x^2+\frac{15}{8\cdot 3!}x^3+R_3 \\ &= 1+\frac{1}{2}x+\frac{3}{8}x^2+\frac{5}{16}x^3+R_3 \end{aligned}$$

where the remainder term is

$R_3(x) = \dfrac{35x^4}{128(1-c)^{9/2}}$, with $0 < |c| < |x|$.

[5]

Most got this right. Errors in differentiation, mostly. A **lot** *of people left unreduced fractions such as $15/48$: bad form, always simplify! Few people gave the remainder term, leaving it instead as $x^4 f^{(4)}(c)/4!$. If the correct derivative $f^{(4)}$ is shown elsewhere, along with the correct inequality linking c and x, this is almost acceptable.*

ii) Find the maximum error incurred in using the series up to the term in x^4 to estimate $\dfrac{1}{\sqrt{0.9}}$. [You can leave the answer in terms of a fraction.]

Solution:

$\frac{1}{\sqrt{0.9}} = f(0.1)$, so the maximum error will be bound by $|R_3(0.1)|$. Given that $\frac{1}{(1-c)^{9/2}}$ is maximal when the denominator is smallest, and $0 < c < 0.1$, then $c = 0.1$ gives the upper bound:

$$|R_3(0.1)| < \frac{35(0.1)^4}{128(0.9)^{9/2}}.$$

[4]

(For information: The error is ≈ 0.00003 and the bound is ≈ 0.00004.)

The question with the least correct answers in Part A. Many put in things like $\sqrt{0.9}$ in the place of x or c; many left x^4 in the remainder, or just dropped it altogether, with varyingly odd results. Many had the right approach but didn't think about where in the interval $[0, 0.1]$ should c be, and chose the simpler $c = 0$.

c) Given the power series $\sum_{n=1}^{\infty} \frac{x^n}{2^n - 1}$, find all values of x for which the series converges.

Solution:

Use the ratio test:

$$\frac{a_{n+1}}{a_n} = \frac{\frac{x^{n+1}}{2^{n+1}-1}}{\frac{x^n}{2^n-1}} = \frac{2^n - 1}{2^{n+1} - 1} x = \frac{1 - 2^{-n}}{2 - 2^{-n}} x,$$

and, as $2^{-n} \to 0$, in the limit, we have

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{1 - 2^{-n}}{2 - 2^{-n}} |x| = \frac{1}{2}|x|.$$

So by the ratio test, the series converges for $\frac{1}{2}|x| < 1 \Rightarrow -2 < x < 2$ and diverges for $\frac{1}{2}|x| > 1 \Rightarrow x > 2$ or $x < -2$. When $\frac{1}{2}|x| = 1$, the ratio test is inconclusive. When $x = 2$ the series diverges; easy to show as all terms a_n are positive and $a_n \to 1$. When $x = -2$, the series converges. To see this, one valid approach is to consider consecutive pairs:

$$\left(-\frac{2}{1} + \frac{4}{3}\right) + \left(-\frac{8}{7} + \frac{16}{15}\right) + \left(-\frac{32}{31} + \frac{64}{63}\right) \cdots = -\frac{2}{3} - \frac{8}{105} - \frac{32}{1953} - \cdots$$

or, in general:

$$\frac{(-2)^n}{2^n - 1} + \frac{(-2)^{n+1}}{2^{n+1} - 1} = \frac{(-2)^n (2^{n+1} - 1) + (2^n - 1)(-2)^{n+1}}{(2^n - 1)(2^{n+1} - 1)} = \frac{(-2)^n}{(2^n - 1)(2^{n+1} - 1)}$$

Now rearrange the sum to $n = 1, 3, 5, 7 \ldots$ and convergence is reasonably easy to demonstrate, for example with the ratio test.

Thus the desired interval of convergence is $[-2, 2)$.

[4]

Most people got convergence on $(-2,2)$ correct. Some couldn't handle the limit of $\frac{1-2^{-n}}{2-2^{-n}}$ and ended up with $(-1,1)$. Not many people gave any thought about what happens for $x=\pm 2$. Given the difficulty of showing convergence for $x=-2$, full credit was given for recognizing that there was an open question.

3. *This is a question on ODEs (Part (a) is a bit unusual in that the solution is given and the students are asked to derive the differential equation, Parts (b) and (c) are on separable and exact ODEs and Part (d) is on linear ODEs, although a transformation is first needed). The students have done well on this question, scoring approximately 69%.*

a) The first two terms of y form the complementary function y_c. Therefore the roots of the auxiliary polynomial $a(\lambda)$ are 1 and 2 so $a(\lambda)=(\lambda-1)(\lambda-2)=\lambda^2-3\lambda+2$ and the LHS of the equation is $\dfrac{d^2y}{dx^2}-3\dfrac{dy}{dx}+2y$. The last term of y is the particular integral y_p and since it is not part of y_c the RHS of the differential equation has the form $f(x)=ce^{3x}$. Substituting $y_p(x)=e^{3x}$ into the LHS, we get $(3)^2-3(3)+2=2$ and so $f(x)=2e^{3x}$. Thus the differential equation is

$$\boxed{\frac{d^2y}{dx^2}-3\frac{dy}{dx}+2y=2e^{3x}} \qquad [\,6\,]$$

b) We recognise the equation as separable by writing the equation in the form

$$\frac{dy}{dx}=\frac{x}{y}e^{x-y}=\frac{x}{y}e^xe^{-y}=(xe^x)\,(ye^y)^{-1}$$

Cross–multiplying, integrating by parts and rearranging the solution is given as

$$ye^ydy=xe^xdx \;\Rightarrow\; \int ye^ydy=\int xe^xdx \Rightarrow ye^y-\int e^ydy=xe^x-\int e^xdx$$
$$\Rightarrow \quad \boxed{(y-1)e^y-(x-1)e^x+C=0} \qquad [\,6\,]$$

c) i) Write the differential equation as $P(x,y)dx+Q(x,y)dy=0$ where $P(x,y)=2xy+e^x$ and $Q(x,y)=x^2+\cos y$. The equation is exact since

$$\boxed{\frac{\partial P}{\partial y}=2x=\frac{\partial Q}{\partial x}} \qquad [\,3\,]$$

ii) Since the equation is exact, the solution has the form $f(x,y)=0$ where

$$\frac{\partial f}{\partial x}=P(x,y)=2xy+e^x, \qquad \frac{\partial f}{\partial y}=Q(x,y)=x^2+\cos y$$

Integrating in turn gives

$$f(x,y)=x^2y+e^x+g_1(y), \qquad f(x,y)=x^2y+\sin y+g_2(x)$$

It follows that $g_1(y)=\sin y$ and $g_2(x)=e^x$ so the general solution is

$$\boxed{x^2y+\sin y+e^x+C=0}$$

[3]

d) i) The transformation $y=\dfrac{dz}{dx}$ gives the linear first order ODE

$$\boxed{\frac{dy}{dx}-\frac{3}{x}y=-3x}$$

[3]

ii) Multiplying by the integrating factor $\mu(x)=e^{\int\frac{-3}{x}dx}=x^{-3}$ gives

$$x^{-3}\frac{dy}{dx}-3x^{-4}y=-3x^{-2}\Rightarrow\frac{d}{dx}\left(x^{-3}y\right)=-3x^{-2}$$
$$\Rightarrow \quad y=x^3\left(\int-3x^{-2}dx+C\right)=x^3\left(3x^{-1}+C\right)=3x^2+Cx^3$$

Finally $z=\int ydx=\int 3x^2+Cx^3dx$ and so $\boxed{z(x)=x^3+C_1x^4+C_2}$ [4]

4. *This question is on functions of two variables. The students did well on this question, scoring an average mark of 66%. Part (a) is on standard partial differentiation, Part (b) is on using the chain rule to solve a symmetric PDE and Part (c) is on obtaining and classifying stationary points using the gradient and Hessian.*

a) i) $\boxed{\dfrac{\partial z}{\partial x} = e^{xy} + y(x+y)e^{xy}}$ [2]

ii) Since $dF = 0$ we have that $\dfrac{\partial F}{\partial x}dx + \dfrac{\partial F}{\partial y}dy + \dfrac{\partial F}{\partial z}dz = 0$. It follows that

$$\frac{\partial z}{\partial x} := \frac{dz}{dx}|_{y=\text{constant}} = \frac{dz}{dx}|_{dy=0} = -\frac{\partial F}{\partial x}\Big/\frac{\partial F}{\partial z} \Rightarrow \boxed{\frac{\partial z}{\partial x} = \frac{z\sin x}{e^z + \cos x}}$$

[3]

iii) Since the integration is with respect to x, $\boxed{\dfrac{\partial z}{\partial x} = \dfrac{xy+1}{x^2+y^3}}$ [3]

b) i) Evaluating the partial derivative for the chain rule

$$\begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} \dfrac{x}{\sqrt{x^2+y^2}} & \dfrac{-y}{x^2+y^2} \\ \dfrac{y}{\sqrt{x^2+y^2}} & \dfrac{x}{x^2+y^2} \end{bmatrix} \begin{bmatrix} \dfrac{\partial f}{\partial \rho} \\ \dfrac{\partial f}{\partial \phi} \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi/\rho \\ \sin\phi & \cos\phi/\rho \end{bmatrix} \begin{bmatrix} \dfrac{\partial f}{\partial \rho} \\ \dfrac{\partial f}{\partial \phi} \end{bmatrix}$$

and so $\left(\dfrac{\partial f}{\partial x}\right)^2 + \left(\dfrac{\partial f}{\partial y}\right)^2$

$$= \begin{bmatrix} \dfrac{\partial f}{\partial \rho} & \dfrac{\partial f}{\partial \phi} \end{bmatrix} \begin{bmatrix} \cos\phi & \sin\phi \\ -\sin\phi/\rho & \cos\phi/\rho \end{bmatrix} \begin{bmatrix} \cos\phi & -\sin\phi/\rho \\ \sin\phi & \cos\phi/\rho \end{bmatrix} \begin{bmatrix} \dfrac{\partial f}{\partial \rho} \\ \dfrac{\partial f}{\partial \phi} \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{\partial f}{\partial \rho} & \dfrac{\partial f}{\partial \phi} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1/\rho^2 \end{bmatrix} \begin{bmatrix} \dfrac{\partial f}{\partial \rho} \\ \dfrac{\partial f}{\partial \phi} \end{bmatrix} = \boxed{\left(\frac{\partial f}{\partial \rho}\right)^2 + \frac{1}{\rho^2}\left(\frac{\partial f}{\partial \phi}\right)^2}$$ [4]

ii) Since $f_\rho = 0$ the equation transforms to

$$\left(\frac{\partial f}{\partial \phi}\right)^2 = \frac{x^2}{(x^2+y^2)} = \cos^2\phi \Rightarrow \frac{\partial f}{\partial \phi} = \pm\cos\phi \Rightarrow f(\phi) = \pm\sin\phi + C$$

and so $\boxed{f(x,y) = \pm\dfrac{y}{\sqrt{x^2+y^2}} + C}$ [4]

c) i) The gradient of f is given by

$$\boxed{\nabla f(x,y) = \begin{bmatrix} f_x \\ f_y \end{bmatrix} = \begin{bmatrix} 6x^2 - 4x - 2y \\ 2(y-x) \end{bmatrix}}$$

The stationary points are obtained as the solutions of the set of equations $\nabla f(x,y) = 0$ and so $\boxed{(x_1,y_1) = (0,0),\ (x_2,y_2) = (1,1)}$ [4]

ii) The Hessian is given by $M(x,y) = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} = \begin{bmatrix} 12x-4 & -2 \\ -2 & 2 \end{bmatrix}$. It follows that $M(0,0) = \begin{bmatrix} -4 & -2 \\ -2 & 2 \end{bmatrix}$ and $M(1,1) = \begin{bmatrix} 8 & -2 \\ -2 & 2 \end{bmatrix}$. A calculation shows that the eigenvalues of $M(0,0)$ have opposite signs so $(0,0)$ is a saddle point and that the eigenvalues of $M(1,1)$ are both positive and so $(1,1)$ is a local minimiser. [5]